

E-COMMERCE SALES ANALYSIS PROJECT

● WHERE EVERY TRANSACTION TELLS A STORY



Dataset

9,994 Orders

Tool (Python)

Pandas + Plotly

Tool (SQL)

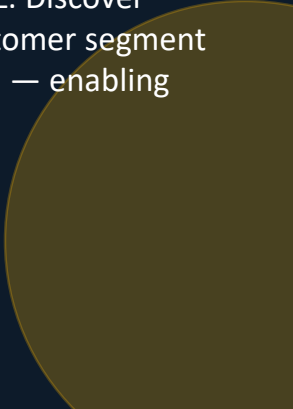
MySQL 8.0

Domain

Superstore Sales

ABOUT PROJECT

Analyze e-commerce sales using Python & SQL: Discover total revenue, top-performing categories, customer segment behaviour, monthly trends, and profit margins — enabling smarter, data-driven business decisions.



DATASET OVERVIEW & DATA CLEANING

```
data = pd.read_csv("Sample - Superstore.csv",  
                  encoding="latin1")  
  
data['Order Date'] = pd.to_datetime(  
    data['Order Date'])  
  
print(data.isnull().sum())
```

Result Grid

Column	Type
Order Date	datetime64
Ship Date	datetime64
Sales	float64
Null Values	0 (Clean)

● 9,994 total records

● 21 columns

● No null values

● 4 years of data

TOTAL REVENUE & PROFIT OVERVIEW

SELECT

ROUND((SUM(profit) / SUM(sales))

* 100, 2)

AS profit_margin_percentage

FROM orders;

total_sales = data['Sales'].sum()

total_profit = data['Profit'].sum()

TOTAL REVENUE

\$2,297,201

Gross Sales

TOTAL PROFIT

\$286,397

Net Profit

PROFIT MARGIN

12.47%

Overall Margin

SALES ANALYSIS BY CATEGORY

```
sales_by_category = data.groupby(  
    'Category')['Sales'].  
    sum().reset_index()  
  
fig = px.pie(sales_by_category,  
    values='Sales', names='Category')
```

Result Grid

Category	Sales (\$)
Technology	836,154
Furniture	741,999
Office Supplies	719,047

Revenue Breakdown

Technology



36.4% \$836K

Furniture



32.3% \$742K

Office Supplies



31.3% \$719K

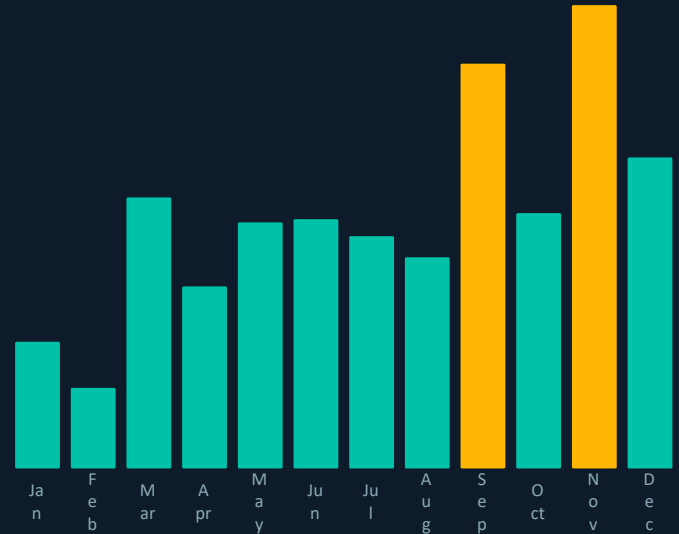
MONTHLY SALES ANALYSIS

```
sales_by_month = data.groupby(  
    'Order Month')['Sales'].  
    sum().reset_index()  
  
fig = px.line(sales_by_month,  
    x='Order Month', y='Sales',  
    title='Monthly Sales Analysis')
```

Result Grid (selected months)

Month	Total Sales (\$)
September (9)	307,649
November (11)	352,461
December (12)	325,294

Monthly Trend



PROFIT ANALYSIS BY SUB-CATEGORY

```
Profit_by_sub = data.groupby('Sub-Category')['Profit'].  
    sum().reset_index()  
  
fig = px.bar(Profit_by_sub,  
    x='Sub-Category',  
    y='Profit',  
    title='Profit By Sub Category')
```

Result Grid

Sub-Category	Profit (\$)
Copiers	55,618
Phones	44,516
Accessories	41,937
Paper	34,054

⚠ Tables (-17,725), Bookcases (-3,473), Supplies (-1,189) = LOSSES

Top 5 Profitable Sub-Categories

Copiers



Phones



Access.



Paper



Binders



SALES VS PROFIT BY CUSTOMER SEGMENT

```
seg_data = data.groupby('Segment').  
    agg({'Sales':'sum',  
        'Profit':'sum'}).reset_index()  
seg_data['ratio'] =  
    seg_data['Sales'] /  
    seg_data['Profit']
```

Result Grid

Segment	Sales (\$)	Ratio
Consumer	1,161,401	8.66
Corporate	706,146	7.68
Home Office	429,653	7.13

Segment Breakdown

Consumer

Sales: \$1.16M

Profit: \$134K

50.6
%

Corporate

Sales: \$706K

Profit: \$92K

30.7
%

Home Office

Sales: \$430K

Profit: \$60K

18.7
%

YEAR-OVER-YEAR SALES GROWTH

```
SELECT year, total_sales,  
  
LAG(total_sales) OVER  
  (ORDER BY year)  
  AS prev_year_sales,  
ROUND((total_sales -  
  LAG(total_sales) OVER  
    (ORDER BY year)) /  
  LAG(total_sales) OVER  
  
    (ORDER BY year)) * 100, 2)  
  AS growth_percentage  
  
FROM (...) t;
```

Result Grid

Year	Total Sales	Growth %
2014	\$484,247	—
2016	\$609,206	+29.46%
2017	\$733,215	+20.36%

YoY Growth Results

2014

Sales: \$484K

2015

Sales: \$470K

-2.83%

2016

Sales: \$609K

+29.46%

2017

Sales: \$733K

+20.36%



THANK YOU

FOR ATTENTION

● 2026 E-COMMERCE SALES
PRESENTATION